

Superstore Sales Analysis Project

From Messy Data to Dashboard

A deeper reflection on what I did in the project and lessons I learned from it.

Project Overview

Preliminary Phase: Raw Dataset Acquisition

Sourcing a dataset suitable for the project.

Phase 1: Data Cleaning

Cleaning the data with Power Query in Microsoft Excel.

Phase 2: Feature Engineering

Building new data from the existing data with Power Query.

Phase 3: Pivot Analysis

Creating PivotTables from the cleaned and augmented data.

Phase 4: Dashboard Development

Creating visualisations (Pivot Charts) from the PivotTables.

Overall Project Reflections

Principal lessons learned from the project.

Potential Next Steps

Ideas for consolidating and enhancing skills developed on the project.

Detailed Review

Preliminary Phase: Raw Dataset Acquisition

This phase covered the initial steps necessary to get to a point where I could begin a meaningful data analysis exercise.

Sourcing An Appropriate Dataset for Project

Actions

I sourced a deliberately messy dataset (containing duplicate rows, inconsistently formatted data, etc.) from kaggle.com, a data science platform. The dataset was called 'Superstore' and contained approximately 10k rows of mock sales data for a typical store. I imported this dataset into Microsoft Excel Power Query, ready for cleaning and transformation.

Takeaway

The project was intended to simulate a more realistic analytical environment. A key aspect was balancing difficulty with manageability.

To make it suitably difficult, the data should not be in an ideal (or close-to-ideal state), but rather in a raw, unreliable state, so that I could learn how to prepare it for trustworthy analysis.

On the other hand, if the dataset was too large or unwieldy, it would make the project excessively long and complex for a first data analysis project aimed at introducing me to the key principles.

The Superstore dataset was a good choice for this project. It is a much longer and more complex dataset than those used for a typical Excel training course exercise on Power Query, but conversely, it is not so complex that it would be overwhelming for me on my first analysis project.

Phase 1: Data Cleaning

This phase was concerned with working towards analytical correctness rather than applying cosmetic formatting.

Generate Clean Dataset

Actions

This step entailed creating multiple queries from the raw dataset, so that the raw data was preserved in its original state, with a 'Working' version to which the cleaning steps were applied. Other queries were created later.

Takeaway

The key lesson here was shifting my thinking about the different queries as copies of the original dataset to thinking of them as different views or perspectives on that same underlying data, each having its own purpose. This made me aware that analytical workflows are modular, reproducible, and non-destructive.

Null and Blank Checks

Actions

This step involved using Power Query's column profiling to assess null and blank values. This tool quickly produces concise but detailed analysis on the data in a particular column.

Takeaway

Using this tool demonstrated clearly to me that Power Query tools are extremely effective in understanding dataset completeness and quality. The tools enable clearer and more sophisticated analysis of the data than manually searching for missing values. I learned that missing values are not automatically errors and that completeness must be understood within the context of the data.

Text Cleaning

Actions

Used the Trim and Clean functions on the text fields to remove hidden spaces, non-printable characters, and inconsistent text formatting (n.b. at this point, *all* fields were text as data type corrections didn't happen until the next step).

Takeaway

I learned that analytical tools rely on consistent underlying representations of data. Standardising the data reduces hidden inconsistencies that may distort grouping, aggregating, filtering, joining, and comparing data.

Data Type Corrections

Actions

This step is concerned with changing any fields that might require it to a more suitable data type. I discovered during this process that decimal format gives full precision, while currency format fixes to 4 decimal places.

Takeaway

Correct data types are crucial for trustworthy analysis, allowing accurate calculation, sorting, and aggregation of data, and the correct functioning of PivotTables.

However, the broader lesson from this is gaining an appreciation that the data cleaning phase in Power Query is more concerned with data correctness; in this case, for

example, the financial data fields are better stored as decimals to enable as much accuracy as possible. Currency format is more visually appealing and easier to read, but the presentational logic should be left to the later Excel analysis phase.

Error Investigation

Action

Power Query has diagnostic filtering tools - in this step I used the Keep Errors tool, which isolates problematic records, i.e. it retains only the error records, making it much easier to see them. Once it had served its purpose, this step was removed from the Applied Steps record.

Takeaway

The key lesson here is that manipulating data with tools to reveal errors is better than visually scanning among the full dataset, trying to find them manually. Another lesson was that diagnostic steps are temporary investigative tools, not permanent transformations of the data.

Duplicates Check

Action

This step was more challenging than I had anticipated. Working through this step, I realised that identifying potential duplicates without disturbing the data is much more difficult with a large dataset than it is with a simple training course type dataset of perhaps 50-100 rows.

The first task was determining a uniqueness key – Order ID and Product ID were sufficient for this. Next, grouping and filtering the data, and then creating a referenced query, which merged the working data with the grouped data to give the fullest picture of the data that might contain duplicates. This allowed me to verify the full row context, and so identify the confirmed duplicates, so that only these were permanently removed.

Takeaway

The major principle here is that unusual repetition of data is not automatically erroneous duplication. Consequently, duplicate handling requires careful judgement.

Rather than doing quick scans to find duplicates and rushing in to remove them, it's far better to do a detailed investigation into the context of the data to determine whether they are duplicates or not, applying business logic, and only remove a record when confident it is a duplicate.

Phase 2: Feature Engineering

This phase was concerned with creating new analytical meaning from existing data.

Data Transformation - Splitting Customer Names

Action

I split the Customer Name field into First Name and Last Name fields. This involved taking care to preserve any multi-part surnames – treating everything after the first space character in the field as the last name.

Takeaway

Although this step was not a rich one in terms of deep analytical work, it did have some bearing on later analysis work. The key lesson here is that atomic fields are easier to analyse and manipulate than combined fields.

Extracting Date Parts

Actions

Order Date is a crucial piece of data; however, raw dates are difficult to analyse directly. Therefore, component parts were extracted from it - Year, Month, and Month Name.

Takeaway

An existing data field can be transformed to enable trend analysis and period-based comparison that would otherwise be much harder to complete. There was a similar principle here to the customer name field: the order date contains several distinct pieces of data, although in this case, the derived fields were additions to the original data field rather than replacements for it.

Calculating Profit Margin

Actions

I created a calculated column, based on profit divided by sales, with added error checking for divide-by-zero errors. This brought the notion of relative profitability to the surface – a useful additional piece of data alongside absolute profit values.

Takeaway

The high-level lesson here was that derived metrics can reveal analytical relationships hidden within raw values. But there was also a further key idea that emerged from applying that principle to this dataset: high sales revenue does not automatically mean the business is making profit *efficiently*.

Profit Status

Actions

Created a calculated column with conditional logic to classify transactions (as Profit, Loss, or Break Even) based on the profit margin. Transactions with negative profit margins were intentionally retained.

Takeaway

This step demonstrated that conditional classifications can simplify complex numerical data into more interpretable analytical categories, improving usability of PivotTables and clarity of Pivot Charts. It also illustrated that unusual values are not automatically invalid values.

Phase 3: Pivot Analysis

This phase shifted the focus from preparing data to developing structured meaning from the data.

Pivot Tables

Sales & Profit by Category

Purpose: Provides revenue/profitability comparison across product categories.

Takeaway: It cannot be assumed that high sales figures automatically mean proportionally high profits.

Profit by Region

Purpose: Compares regional profitability over time.

Takeaway: Breaking down data geographically can reveal performance patterns across those geographical areas.

Profit Impact by Profit Status

Purpose: Shows sales/profit with their profitability status. Evolved to become a “master” analytical table, with specific tables derived from it that show *Sales by Profit Status* and *Profit by Profit Status*.

Takeaway: Complex analytical structures may support insight generation better than direct visual presentation.

Transaction Count by Profit Status

Purpose: Provides operational context alongside the financial metrics.

Takeaway: Considering financial impact alone does not provide a full understanding. Adding the frequency element illuminates the inquiry, distinguishing between isolated high-impact losses and patterns of widespread low margins.

Key Takeaways

Analytical structure should follow the business question

Effectively exploring profitability, regional performance, and transaction frequency required different PivotTables, aggregation levels, and metrics.

Financial metrics require operational context

Transaction counts complemented sales/profit figures, revealing the frequency of loss-making transactions.

PivotTables support visual communication

PivotTables often functioned as intermediate analytical structures used to create clearer charts downstream (in the dashboard phase) rather than as financial presentation outputs in themselves.

Phase 4: Dashboard Development

This phase evolved from an initial simplistic approach:

Learning to make charts

into a more sophisticated one:

Examining what a chart is attempting to communicate.

Key Takeaways

Dashboard development is iterative

Creating effective visuals requires repeatedly refining both PivotTables and Pivot charts to improve clarity and focus.

Simpler visuals often communicate insights more effectively.

Separating metrics and reducing chart complexity improved the points the chart was intended to illustrate.

Visual design should reinforce the analytical question

Chart structure, metrics, and formatting were adjusted to emphasise the trends most relevant to each analysis objective.

Overall Project Reflections

The principal lesson comes from the transformation of my mindset on how to approach a project of this nature: initially I approached it in terms of following procedural steps in the software, but it evolved into understanding analytical purpose.

Prior to undertaking this project, I had a basic understanding of the processes in Excel to import data with Power Query and creating PivotTables and Pivot Charts having completed exercises on these topics within advanced Microsoft Excel training courses. Therefore, I went into it thinking very much in terms of learning the tools within the software and wanting to improve my understanding of them, largely to produce end results of charts and tables, without really understanding why (and how) those end results might be useful.

However, during the various phases of the project, my mindset shifted towards understanding the analytical purpose of each stage of the workflow. Although this exercise revolved around a contrived set of data purely for learning purposes, I tried to imagine how real business users might approach the data and what they might want out of it.

If I came to a point where I wasn't sure what the reasoning behind a suggestion was, or what a particular task was intended to do, I would "press" on it to gain a deeper understanding and sometimes to alter the plan to take a more relevant course of action.

Major Themes Learned

Analytical Correctness Versus Presentation

The key purpose of the data cleaning phase is to prepare the data to be ready for robust analysis in the later phases, not simply to making it look neat on the screen.

Systematic Interpretation of Data

Using structured, scalable methods to interrogate the dataset is a more robust and effective approach than relying on quickly visually scanning the data and making educated guesses about it.

Contextual interpretation of Data

Striving to understand what the data means in a potentially genuine business context is more effective than a simplistic training exercise style evaluation.

Iterative Refinement of Analysis

The most effective and meaningful analytical outputs usually result from cyclical improvements (as a better understanding of the data and its potential meanings develops) rather than being created directly on the first attempt.

Communication Clarity

Visuals can be technically correct and impressive but not necessarily communicate key ideas well; the most valuable analysis communicates insights clearly, allowing them to be easily interpreted.

Purpose-Driven Analysis

Effective analysis is not about picking the most attractive or impressive charts. Instead, a more sophisticated approach is required, in which the analytical structure and visual design is guided by business questions or insights being investigated.

Potential Next Steps

Completing this project, I developed a strong foundational understanding of:

- Business Intelligence workflows
- Structured analytical reasoning
- Dashboard design and visual communication

To build on this foundation, potential next steps I could take include:

- Extending dashboarding and visualisation skills in Power BI
- Exploring more advanced analytical modelling and interactivity
- Applying the same structured workflow to larger and messier datasets
- Improving data storytelling and presentation skills